25901229 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECT200 - ARM BASED EMBEDDED SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Register allocation in Embedded C refers to [U] / [CO1]

   a). Allocating memory in heap

   b). Allocating I/O ports

   c). Assigning registers to peripherals

   d). Mapping variables to CPU registers

2. Which feature is unique to AXI? [U] / [CO2]

   a). Low power b). Split transactions

   c). Out-of-order completion d). Single clock edge

3. In ARM-based systems, device drivers are generally: [U] / [CO3]

a). Hardware-dependent b). Hardware-independent

c). Written only in assembly language d). Part of application software

4. Which function is used to configure a GPIO pin as OUTPUT in ARM/Arduino? [An] / [CO4]

a). digitalWrite() b). pinMode() c). delay() d). analogRead()

5. Which instruction is used for conditional branching in ARM? [K] / [CO1]

a). BNE b). CMP c). MUL d). MOV

6. Pull-up/down resistors are used to: [An] / [CO4]

a). Increase current b). Avoid floating input c). Blink LED d). Trigger interrupt

7. Which register acts as the program counter? [K] / [CO2]

a). APSR b). R15 c). R12 d). R10

8. SPI communication is generally: [U] / [CO3]

a). Multi-master only b). Single-master multi-slave

c). Single-slave only d). Broadcast

9. Which of the following is a specialized ARM processor? [K] / [CO1]

a). ARM7TDMI b). ARM9E

c). ARM Cortex-M d). ARM11 MPCore

10. ARM Cortex-M processors are mainly intended for [U] / [CO2]

a). Mobile phones

b). High-performance computing

c). Microcontroller applications

d). Server systems

11. In ARM Cortex-M microcontrollers, GPIO ports are usually accessed via [U] / [CO3]

a). Stack memory

b). Special Function Registers (SFRs)

c). Cache memory

d). EEPROM

12. Which of the following modes is privileged in ARM 7? [U] / [CO2]

a). System

b). User

c). User and System

d). None

13. Output compare can be used to: [U] / [CO3]

a). Toggle a pin

b). Set a pin

c). Clear a pin

d). All of the above

14. Which interrupt has higher priority in ARM? [K] / [CO2]

a). FIQ

b). IRQ

c). Reset

d). Software Interrupt

15. Harvard architecture requires [U] / [CO1]

a). Single memory unit

b). More hardware and control logic

c). Less hardware complexity

d). No control unit

16. Which ARM feature improves instruction throughput? [K] / [CO1]

a).Microprogramming

b). Polling

c). Pipelining

d). Segmentation

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. Compare IRQ and FIQ exceptions with respect to latency and register usage. [U] / [CO2]

18. List and discuss the design principles of Embedded System. [U] / [CO3]

19. Describe the program logic of a buzzer and LED alert system triggered by sensor input. [An] / [CO4]

20. Explain the difference between CPSR and SPSR. [U] / [CO2]

21. What is register allocation in embedded C for ARM? [K] / [CO1]

22. Explain the Cortex-M memory map [K] / [CO2]

23. Compare ODR and IDR registers in ARM Cortex-M GPIO. [U] / [CO3]

24. What is the use of the `volatile` keyword in Embedded C? [K] / [CO1]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. With a neat diagram describe an Embedded System. Explain the classifications based on functionality and performance. [K] / [CO1]

26. With the help of flow diagram describe the ARM exception handling mechanism. Explain the complete sequence of events during exception entry and exception return. [U] / [CO2]

27. Explain the working principle and interfacing of an IR sensor with an ARM Cortex-M microcontroller. [U] / [CO3]

28. Explain how a Seven-segment Display may be interfaced to an ARM based microcontroller? [An] / [CO4]

[12x2 = 24]